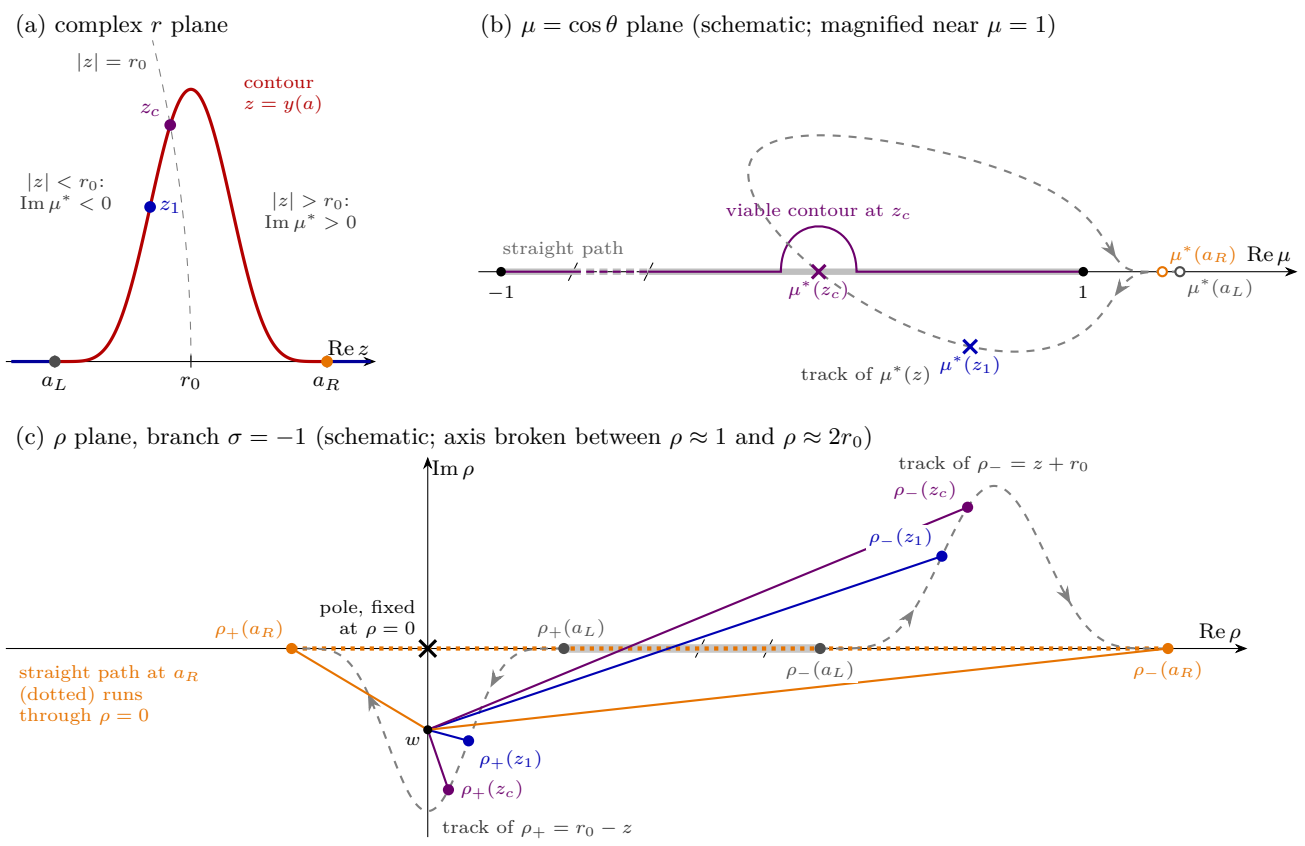

(a) complex r plane
$|z| = r_0$
contour
$z = y(a)$
z_c
$|z| < r_0$:
$\mathrm{Im}\,\mu^* < 0$
z_1
$|z| > r_0$:
$\mathrm{Im}\,\mu^* > 0$
$\mathrm{Re}\,z$
a_L
r_0
a_R
(b) $\mu = \cos\theta$ plane (schematic; magnified near $\mu = 1$)
viable contour at z_c
straight path
$\mu^*(a_R)$
$\mathrm{Re}\,\mu$
$\mu^*(a_L)$
-1
$\mu^*(z_c)$
1
track of $\mu^*(z)$
$\mu^*(z_1)$
(c) ρ plane, branch $\sigma = -1$ (schematic; axis broken between $\rho \approx 1$ and $\rho \approx 2r_0$)
$\mathrm{Im}\,\rho$
track of $\rho_- = z + r_0$
$\rho_-(z_c)$
$\rho_-(z_1)$
pole, fixed
at $\rho = 0$
$\rho_+(a_R)$
$\rho_+(a_L)$
$\mathrm{Re}\,\rho$
straight path at a_R
(dotted) runs
through $\rho = 0$
$\rho_-(a_L)$
$\rho_-(a_R)$
w
$\rho_+(z_1)$
$\rho_+(z_c)$
track of $\rho_+ = r_0 - z$